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INTRODUCTION

Chronic kidney disease (CKD) is a progressive condition characterized by structural or functional abnormalities of the kidney with or without a low rate of glomerular filtration^{1,2}. Hypertension, cardiovascular disease and diabetes mellitus are the main etiological factors and cardiovascular disease is described as the main cause of death in individuals with CKD³⁻⁶. The prevalence of CKD has increased significantly, with 13.4% of the population affected worldwide⁷.

The treatment for CKD is based on renal replacement therapy. Dialysis is a process of artificial filtration that removes all undesired substances accumulated due to chronic renal failure. Hemodialysis is the most common form of therapy for such patients².

Individuals with CKD who are undergoing hemodialysis are likely to have oral problems and inadequate oral hygiene. The literature states that poor oral health can contribute to systemic inflammation and atherosclerotic complications, leading to a higher risk of mortality⁸⁻¹⁰. The most frequent oral problems in individuals with CKD are a reduction in salivary flow, a higher prevalence rate of dental caries and periodontal disease, xerostomia, candidiasis and a burning sensation in the mouth. Premature tooth loss and edentulism are also common in individuals with this illness¹¹⁻¹⁴.

A recent systematic review reports that approximately one in five individuals with CKD are edentulous¹³. The main etiology and pathogenesis of edentulism in such cases is severe periodontal disease¹⁵. Besides clinical parameters of edentulism, it is also important to analyze the social and individual determinants involved in the occurrence of this condition. Disparities in oral health related to race and socioeconomic status are also reported to be strong determinants of tooth loss^{16,17}.

The aim of the present study was to identify risk factors associated with edentulism in adults with CKD undergoing hemodialysis. Based on the literature, the authors tested the hypothesis that, controlling for age, edentulous individuals with CKD undergoing hemodialysis have a greater chance of being women, having low schooling, having undergone hemodialysis for a longer period of time, having other diseases associated with CDK, having not visited a dentist in the previous six months, having a negative perception of their smile and having alterations in the oral mucosa.

METHODS

Ethical aspects

e with CKD and undergoing hemodialysis during the data collection period. re recruited from two hemodialysis centers located in the cities of Belo Horizonte, in the southeast of Brazil. Data collection took place September and November 2011.

Eligibility criteria

Training and calibration process

A calibration process and pilot study were conducted prior to the main study. The training and calibration process consisted of theoretical and clinical steps^{18,19}. Discussions were conducted by an experienced dentist using slides for the diagnosis of alterations in the oral mucosa. During the clinical step of the calibration process, thirteen individuals were examined and reexamined after a seven-day interval. Examinations were performed using a portable artificial light (petzl headlamp) and a disposable mouth mirror (Prisma, São Paulo, SP, Brazil) using the criteria established by the World Health Organization²⁰. These individuals were from the hemodialysis center in Belo Horizonte and did not participate in the main study. The kappa statistic was used for the determination of inter-examiner ($k = 0.86$) and intra-examiner ($k = 0.90$) agreement regarding the diagnosis of mucosal alterations. The kappa values were considered good¹⁹.

A pilot study was also conducted to test the methods and data collection instruments. For such, a convenience sample was used of 60 individuals with CKD treated at the same hemodialysis centers. The results of the pilot study demonstrated no need to alter the methods and these individuals were included in the main study.

Nonclinical data collection

Nonclinical data were collected using a structured questionnaire administered in interview form. A single researcher was responsible for applying the questionnaire to the volunteers. The questionnaire was composed of items related to the individual's general characteristics (such as the consumption of alcohol and cigarettes), medical and dental history. Ethnicity was categorized based on the criteria established by the Brazilian Institute of Geography and Statistics (IBGE)²¹. Socioeconomic status was determined based on the Brazilian Economic Classification System²², which categorizes families into classes A (highest), B, C, D and E (lowest). For the purposes of statistical analysis, economic classes were categorized as follows: Group 1 – composed of economically more favored individuals (Classes A and B); Group 2 – composed of economically favored individuals (Class C); and Group 3 – composed of economically less favored individuals (Class D). No patients were categorized in Class E. Schooling was evaluated based on the number of years of formal education.

ty of the questionnaire, testing and retesting of the measure were conducted lunteers who participated in the calibration exercise. The retest was carried en-day interval. Kappa coefficients regarding test/retest agreement ranged from 0.72 to 1.00, which are considered good to excellent, demonstrating an appropriate degree of reliability¹⁹.

Clinical data collection

The oral examinations were performed to evaluate alterations in the oral mucosa using the criteria established by the World Health Organization²⁰. Examinations of the mucosa, palate and tongue (edges, dorsum and base) were performed using a portable artificial light (petzl headlamp) and a disposable mouth mirror (Prisma, São Paulo, SP, Brazil). In cases of food debris and the presence of plaque, the teeth were cleaned with gauze. Denture wearers were instructed to remove the prosthesis prior to the examination. All examinations were performed by a single researcher with the aid of a research assistant who recorded the data, with the individual seated in a reclining chair while undergoing hemodialysis. Individuals with dental treatment needs were referred to the School of Dentistry of the university.

Statistical analysis

The data were analyzed using the Statistical Package for the Social Sciences (SPSS for Windows, version 21.0, SPSS IBM Corp., Armonk, NY, USA). Descriptive and bivariate analyses were performed. The chi-square test was used to determine associations ($p < 0.05$) between the dependent variable (edentulism) and independent variables (age, sex, skin color, socioeconomic class, educational level, duration of hemodialysis, occurrence of other diseases, dental care in the previous six months, self-rated smile and alterations in the oral mucosa). Multiple logistic regression analysis was performed to identify the independent impact of each variable studied. The independent variables were incorporated into the decreasing logistic model in accordance with their statistical significance ($p < 0.25$; stepwise backward procedure).

RESULTS

The final sample comprised 650 individuals with CKD undergoing hemodialysis, representing 84.7% of the universe of individuals in treatment at the selected reference centers. The 117 dropouts were due to absences on the day scheduled for data collection, refusals to participate in the study, participation in the calibration exercise, the occurrence of deaths or other reasons.

Table 1 displays the demographic and socioeconomic characteristics of the sample. Age ranged from 18 to 90 years (mean: 53.9 ± 14.3 years). The majority was male (55.2%), black or brown (80.2%), from the more favored and favored economic classes (84.0%) and had less than eight years of schooling (72.3%).

	Absolute frequency (n)	Relative frequency (%)
18-45	166	25.5
46-65	341	52.5
66-90	143	22.0
Sex		
Male	359	55.2
Female	291	44.8
Skin color/Race		
White	129	19.8
Black/Brown	521	80.2
Yellow/Indigenous	0	0
Economic class		
Most favored	106	16.3
Favored	440	67.7
Less favored	104	26.0
Schooling (years study)		
< 8	470	72.3
≥ 8	180	27.7

The prevalence of complete edentulism was 28.2% (n = 183) and 71.8% of the sample had at least one tooth in the oral cavity (n = 467). Among the edentulous patients, 76.5% wore removable dentures (n = 140).

Table 2 displays the associations between the dependent and independent variables. Statistically significant associations were found between edentulism and age, sex, skin color, economic class and schooling ($p < 0.05$). Moreover, 31.4% of the edentulous individuals stated not having been to a dentist in the previous six months ($p < 0.001$); 37.3% rated their smile as excellent or good ($p < 0.001$) and 43.4% were diagnosed with some mucosal alteration ($p < 0.001$).

		Edentulism			P value**
Variables		Present n (%)	Absent n (%)	Total n (100%)	
Age (years)*	18-45	6 (3.6)	160 (96.4)	166	< 0.001
	46 -65	97 (28.4)	244 (71.6)	341	
	66 -90	80 (55.9)	63 (44.1)	143	
Sex	Female	95 (32.6)	196 (67.4)	291	0.02
	Male	88 (24.5)	271 (75.5)	359	
Skin color	White	48 (37.2)	81 (62.8)	129	0.01
	Black/Brown	135 (25.9)	386 (74.1)	521	
Economic class	More favored	23 (21.7)	83 (78.3)	106	0.02
	Favored	122 (27.7)	318 (72.3)	440	
	Less favored	38 (36.5)	66 (63.5)	104	
Schooling (years)	< 8	163 (34.7)	307 (65.3)	470	< 0.001
	8 or more	20 (11.1)	160 (88.9)	180	
Hemodialysis treatment time (years)	< 5	128 (29.4)	308 (70.6)	436	0.33
	5 or more	55 (25.7)	159 (74.3)	214	
Having another disease	Yes	157 (29.4)	377 (70.6)	534	0.13
	No	26 (22.4)	90 (77.6)	116	
Dental care (< 6 months)	No	154 (31.4)	337 (68.6)	491	< 0.001
	Yes	29 (18.2)	130 (81.8)	159	
Smile perception	Excellent/ Good	79 (37.3)	133 (62.7)	212	< 0.001
	Regular/Poor	104 (23.7)	334 (76.3)	438	
Oral mucosa alteration	Present	121 (43.4)	158 (56.6)	279	< 0.001
	Absent	62 (16.7)	309 (83.3)	371	

* Categorized into tercis ** X² Test (significance level of 5%/ value in bold: statistical significance (< 0.05)).

Table 3 displays the results of the multiple logistic regression analysis. Individuals with less than eight years of study had a 3.99-fold (95% CI: 2.34 to 6.79) greater chance of being edentulous. Individuals who had not been to a dentist in the previous six months had a 2.49-fold (95% CI: 1.52 to 4.08) greater chance of being edentulous. Individuals with an excellent or good perception of their smile had a 2.00-fold (95% CI: 1.35 to 2.97) greater chance of being edentulous. Individuals with alterations identified in the oral mucosa had a 4.17-fold (95% CI: 2.83 to 6.13) greater chance of being edentulous.

variables (Categories)	OR (CI 95%) Crude	OR (CI 95%) Adjusted*
years)	4.24 (2.57-7.01)	3.99 (2.34-6.79)
Dental care < 6 months (absent)	2.04 (1.31-3.19)	2.49 (1.52-4.08)
Smile perception (Excellent/ Good)	1.90 (1.33-2.72)	2.00 (1.35-2.97)
Oral mucosa alteration (Present)	3.81 (2.66-5.47)	4.17 (2.83-6.13)

*Adjusted for control variable (age) / OR: Odds Ratio / CI 95%: Confidence interval.

DISCUSSION

Due to the high levels of psychological stress and lack of time to perform dental care, individuals undergoing hemodialysis often consider oral health maintenance to be a low priority. Researchers report that individuals with psychological stress are more likely to exhibit risk behaviors related to oral health, which leads to a significant increase in the risk of tooth loss^{9,24,25}.

In the present study, the prevalence of edentulism among patients undergoing hemodialysis was 28.2%. In most cases, tooth loss is explained by the high prevalence of periodontal disease and dental caries¹¹⁻¹³. According to a systematic review conducted in 2014, one out of every five adults with CKD is edentulous¹³. The present results are also in agreement with data described in a study conducted in Germany in 2012 with individuals aged 32 to 86 years undergoing hemodialysis, in which the prevalence of edentulism was 22%²⁶.

The findings partially support the hypothesis posed in the present study. Controlling for age, the prevalence of edentulism was associated with most of the variables analyzed (schooling, dental care, self-rated smile and alterations in the oral mucosa). However, the direct association with “self-rated perception” was different from what was expected. Moreover, the hypothesis was not supported with regard to sex, duration of hemodialysis and the occurrence of other diseases associated with kidney failure.

Individuals with less schooling undergoing hemodialysis had a greater chance of being edentulous. This finding may be related to the fact that most individuals with less schooling also have a lower income and probably have less access to dental services as well as fewer financial resources for periodic dental appointments and treatment. The literature shows that low income and schooling are directly related to tooth loss^{4,16,17,27}. A study conducted in the United States in 2014⁴ involving 2749 individuals with CKD found that most of the toothless individuals had a low income and less than 12 years of schooling. Indeed, people with lower schooling generally have the poorest oral health indexes and often only seek treatment when a problem is apparent^{17,27}.

The considerable regional inequalities in Brazil exert a direct influence on access to specialized oral health services. Moreover, even if specialized procedures enable the greater preservation of teeth, tooth extraction is considered inevitable in cases of tooth decay in advanced stages of tissue destruction²⁷. Individuals often only visit a dentist in the occurrence of pain and may prefer to undergo tooth extraction, which they consider to be more practical and less costly. In many cases, individuals are unaware of the future consequences

als undergoing CKD who reported not having been to a dentist in the previous d a 2.5-fold greater chance of being edentulous. This result may be explained at most edentulous individuals don't think about possible oral problems and don't visit the dentist very often. Moreover, individuals use to associate the presence of teeth in mouth with dental care utilization²². Studies have demonstrated that individuals undergoing dialysis only visit a dentist in the occurrence of pain or when the need for treatment becomes evident^{24,26}.

Another point that may be associated with the low frequency of visits to a dental office among this group of patients is the refusal on the part of dentists to provide treatment. According to Jain et al.¹², many dentists refuse care with the justification of not having sufficient knowledge on CKD and due to the fear of some systemic complication. According to the authors, this situation contributes to the considerable need for prosthetic rehabilitation among patients undergoing hemodialysis. A study conducted in China in 2014 revealed that 19.5% of 306 hemodialysis patients did not receive dental treatment due to rejections on the part of dentists¹⁰.

The majority of individuals with CKD in the present study rated their smile as either good or excellent. According to Rodakowska et al.²⁹, oral problems are not considered a major concern among individuals undergoing hemodialysis, which may be due to an adaptation to impaired oral health or a change in health priorities. The present findings also showed that most edentulous individuals made use of complete removable dentures and it is possible that the patients consider their dentures to be more attractive than their previous compromised teeth. In general, dentures have aligned, larger, whiter teeth and provide greater volume to the lips, generating a younger, more pleasing appearance. According to Silva et al.³⁰, complete dentures in edentulous individuals provide patient satisfaction and restore one's self-esteem.

The association between edentulism and the positive perception of one's smile could also be related to the absence of pain in such individuals. Complete extractions often represent a solution for ending pain and it is probable that individuals prefer being without teeth to experiencing pain³¹. Unfortunately, there is also the cultural belief that losing one's teeth over the years is a normal occurrence³².

Individuals with CKD in whom mucosal alterations were identified had a greater chance of being edentulous. It may be related to the use of inadequate or ill-adapted dentures causing injuries to the soft tissues. Also, the literature describes that the presence of mucosal alterations may be associated with insufficient and improper hygiene habits as well as the use of dentures for a long period of time^{33,34}.

Some limitations of the present study should be pointed out. The cross-section design does not enable the evaluation of causal relationships. A non-probabilistic sample was used due to the difficulty in selecting eligible participants, which does not ensure the external validity of the results. The lack of a control group makes it difficult to perform generalizations to other populations. Furthermore, memory bias may have occurred among the participants during the administration of the questionnaire regarding the duration of hemodialysis and the time of the last visit to a dentist.

Tooth loss may result in changes in the stomatognathic system, such as temporomandibular disorder, excessive load on the remaining teeth and difficulty chewing^{35,36}. It can change one's lifestyle by affecting facial esthetics and speech, which exerts a negative impact on social and professional aspects, causing psychosocial problems and reducing one's quality of life^{8,9}.

CKD may lead individuals more susceptible to infections due a reduced immunological response and general health debilitation. On top of that, the investigation performed in the

and changes in the stomatognathic system^{35,36}. Thus, those individuals face health problems that might compromise their general health and quality of life. Evaluation of factors associated with edentulism in CKD individuals can assist in the establishment of better oral health public policies that take into account their specific needs. The high prevalence of edentulism found in the present study underscores the need for adequate dental care to ensure oral health and avoid the undesirable consequences of tooth loss in this group of individuals.

CONCLUSION

In the present study, 28.2% of the individuals with chronic kidney disease undergoing hemodialysis were edentulous. Edentulism was associated with low schooling, a lack of dental care in the previous six months, a positive self-perception of one's smile and alterations in the oral mucosa.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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